David Kieu

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Professional Summary

Data engineer who turns messy real-world data into reliable, tested pipelines and queryable stores. Built and validated automated data and ML pipelines (Python, SQL, PostgreSQL) and a computer-vision data pipeline for document images in a commercial setting, with strong SQL, data-validation, and CI/Docker discipline. Bachelor of Computer Science (AI major, GPA 6.375/7.0) from the University of Adelaide, graduated 2026, AWS Certified Cloud Practitioner, based in Adelaide and open to relocation across Australia.

Technical Skills

Languages: Python, SQL, Bash, JavaScript

Databases & Formats: PostgreSQL, DuckDB, SQL Server, Parquet

Data Engineering: ETL & Data Pipelines, dbt, Dimensional & Schema Modelling, Data Quality Testing, Idempotent / Reproducible Builds

Automation & Tooling: Python Scripting, REST APIs, Task Scheduling, Make

DevOps & Cloud: Docker, AWS, CI/CD, Git, Linux

Education

University of Adelaide — **Adelaide, South Australia**
Bachelor of Computer Science – Artificial Intelligence — *Graduated 2026*
GPA: 6.375/7.0 (completed)

AWS Certified Cloud Practitioner — **December 2025**
Amazon Web Services – Credential ID: AWS05628495

Work Experience

University of Adelaide — **November 2024 – Present**
Research Intern – Machine Learning — *Adelaide, Australia*

- Built a reproducible Python experimentation pipeline (PyTorch) that ran, logged, and validated results across many model configurations, replacing manual, ad-hoc setup with scripted, re-runnable data workflows
- Designed an automated training-and-evaluation pipeline with a curriculum-learning strategy that reduced convergence time, producing systematic, validated outputs documented for research publication

Kaopiz Software JSC — **November 2023 – February 2024**
Machine Learning Engineer

- Built a Python and OpenCV image-processing pipeline to localise Japanese characters across a document-image dataset, scripting reproducible preprocessing from raw images to extracted character regions
- Developed reusable data-preparation scripts for the team's document-recognition R&D, replacing manual, ad-hoc image handling with repeatable processing

Ecosmartvietnam — **September 2022 – January 2023**
Software Engineer

- Architected and shipped a commercial full-stack application (Flask, PostgreSQL) with a normalised relational data model, owning the full SDLC end to end

Projects

Autonomous Multi-Agent Data-Engineering Swarm — **Multi-agent orchestration, Claude Code subagents, Python**

- Built an autonomous multi-agent system that drives a data-engineering task end-to-end through role-specialised subagents (Planner, Builder, Verifier) as a plan-build-verify loop that returns a machine-checkable pass/fail.
- Reliability comes from agent design, not prompting – each role has restricted tools (the Planner cannot edit code, the Verifier cannot write at all), and a file-based data contract is the sole hand-off between stages, so correctness is verifiable rather than left to the model.

ROS-Bag Telemetry Analytics Pipeline — **Python, DuckDB, dbt, Parquet**

- Built a medallion-architecture pipeline that ingests ROS bag recordings of robot sensor streams (IMU, LiDAR, odometry) into Parquet and transforms them through bronze, silver, and gold layers in DuckDB and dbt to expose clean queryable tables
- Automated the full pipeline with a Makefile so a single make command rebuilds the warehouse end-to-end from raw bags, enforcing data quality with dbt tests and an idempotent build that reprocesses any bag into identical, session-partitioned gold tables

Voice Debrief – Agentic LLM Journaling App — **Next.js, TypeScript, PostgreSQL, Drizzle ORM, Zod**

- Built a PostgreSQL-backed system that turns unstructured spoken input into typed, schema-validated rows (Zod), with an editable document that writes corrections back through the data layer rather than around it
- Enforced integrity at the write boundary – schema-validated inserts with retry loops so malformed records cannot enter the store, and deterministic completion criteria instead of model heuristics

Restaurant Management Platform **C#, ASP.NET Core, Dapper, PostgreSQL, React, WebSockets**

- Modelled a normalised PostgreSQL schema for menu, orders, and users and implemented data access with Dapper and parameterised raw SQL, with constraints and indexes backing the hot read paths
- Routed real-time order data to clients over WebSockets and persisted third-party events (Stripe payments, Lightspeed POS sync) consistently alongside order records